

K.R.L Seminar Series

6-March 2026

The lab



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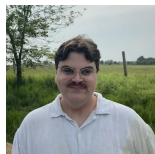
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Seminar 3, Workshop 1

Causal Inference and Difference-in-Differences (DiD)

Outline

- ▶ Understanding Causal Inference
- ▶ Difference-in-Differences (DiD)
- ▶ Workshop
- ▶ Conclusion & Future Steps

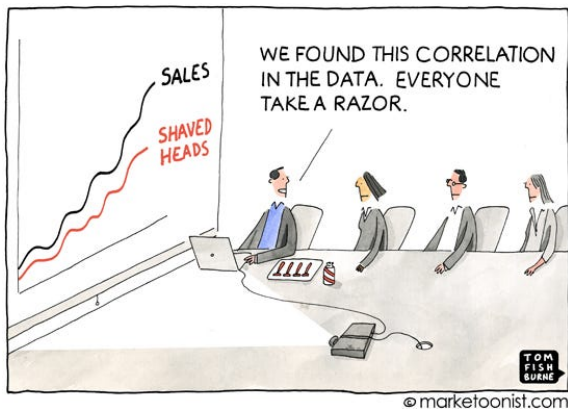
The Questions We Ask in Economics

Examples

- ▶ Did raising the minimum wage reduce employment?
- ▶ Did rent control lower housing supply?
- ▶ Do taxes on cigarettes reduce smoking?

But how do we know these policies caused the change?

Introducing Causal Inference



Source: Dandamudi (2020)

- ▶ **Correlation:** Two variables moving together
- ▶ **Causation:** Identifying whether one variable actually causes changes in another

What is Causal Inference?

From "What Happened?" to "What Caused it?"

- ▶ Outcomes change after a policy, but did the policy itself cause the change?
- ▶ We observe one world – the world with the policy, not the world without it
- ▶ Causal inference constructs that missing comparison (the counterfactual)
- ▶ In policy research like ours, this determines whether we identify **real effects** or just **correlations**

Why Correlation is Not Enough?

- ▶ Just because two things move together, doesn't mean one caused the other
- ▶ Trends, selection, and underlying differences can drive both

For example:

- ▶ Spending on menstrual products may rise due to increase in access to supermarkets – **not tax removal**
- ▶ Shelter access may change due to rising rents – **not Bill 6**
- ▶ Turnout to our seminar may rise – **not because we're giving out free A's in your econometrics classes**

So without a strategy to isolate the policy effect, we risk attributing changes to the wrong cause

Why Does Causality Matter For Our Specific Work?

- ▶ Empirical research relies on causal evidence
- ▶ Correlation can mislead researchers
- ▶ Misattributing effects can lead to biased or misleading conclusions
- ▶ In our projects, this determines whether we identify true treatment effects

So how do we actually construct that counterfactual in practice?

Difference-in-Differences (DiD)

When to use DiD:

DiD is one of many causal inference strategies, and there are plenty of alternatives. Use DiD when:

- ▶ You have **data on outcomes over time** (panel or repeated cross-sections).
 - ▶ **Panel:** Observations of multiple variables from the same group over multiple time periods
 - ▶ **Repeated cross-section:** Observations of multiple variables from *different* groups over multiple time periods¹

¹One would typically need to aggregate repeated cross-section data to a level at which we have repeated observations for the same group: for example, aggregating survey data to the state level...

DiD: The Basics

Classic Example: Card and Krueger (1994).

- ▶ **Policy:** New Jersey raised its minimum wage in April 1992 — to the highest in the country at the time.
- ▶ **Question:** How does a minimum wage increase affect employment in low-wage industries?
- ▶ **Group of interest:** Fast food restaurants in New Jersey.

Problem: What is the counterfactual? How do we know what would have happened in New Jersey *if* the minimum wage hadn't increased?

DiD: The Big Idea

Idea:

Find a group that is **similar** to our group of interest, but that **did not** experience the policy change — and use it to construct the counterfactual.

Card & Krueger's solution

Pennsylvania fast food restaurants did not face a minimum wage hike, but are otherwise comparable to those in New Jersey.

DiD: Treatment & Control Groups

We need:

- ▶ A **treatment group** that experiences the policy change.
- ▶ A **control group** that does not, but **is otherwise similar**.
- ▶ Data on outcomes for both groups **before and after** the policy change.

DiD: Notation

Let $\bar{Y}_{g,t}$ be the sample mean for group g at time t .

- ▶ $\bar{Y}_{T,post}$ is the average value in the treated group after the treatment.
- ▶ $\bar{Y}_{C,pre}$ is the average value in the control group after the treatment.

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Causal inference notation:

- ▶ **ATE**: Average treatment effect

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 - ▶ Huh? Treatment effect on untreated?

DiD: The Estimator

DiD Estimator:

$$\hat{\delta}^{ATT} = \widehat{ATT} = \underbrace{(\bar{Y}_{T,\text{post}} - \bar{Y}_{T,\text{pre}})}_{\Delta \text{ treatment group}} - \underbrace{(\bar{Y}_{C,\text{post}} - \bar{Y}_{C,\text{pre}})}_{\Delta \text{ control group}}$$

- Gives the **Average Treatment Effect on the Treated (ATT)**.

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- Gives the **Average Treatment Effect on the Treated (ATT)**.

	Before (Feb 92)	After (Nov 92)	Change
NJ (Treatment)	20.44	21.03	+0.59
PA (Control)	23.33	21.17	-2.16
DiD Estimate			+2.76

Source: Card & Krueger (1994).

DiD: Graphically

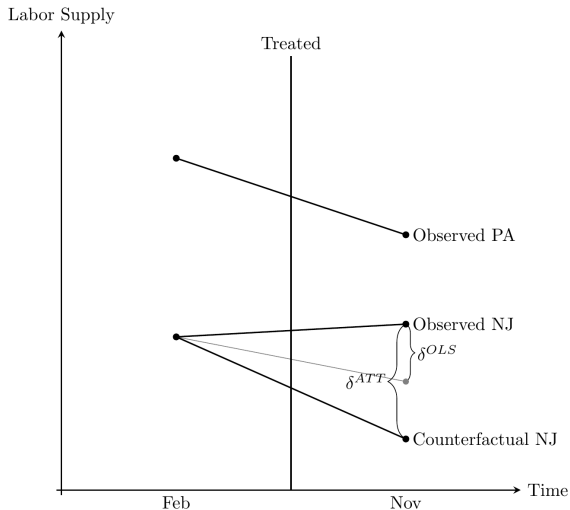


Figure: Difference-in-Differences: Card and Krueger (1994)

Source: Cunningham — Causal Inference Mixtape

DiD: Parallel Trends

Key Assumption — Parallel Trends: In the absence of treatment, the average change in outcomes **would have been the same** for both groups.

- ▶ Any post-treatment *divergence* in trends is attributed entirely to the policy.
- ▶ In our example: absent the minimum wage hike, NJ employment trends would have mirrored PA's.

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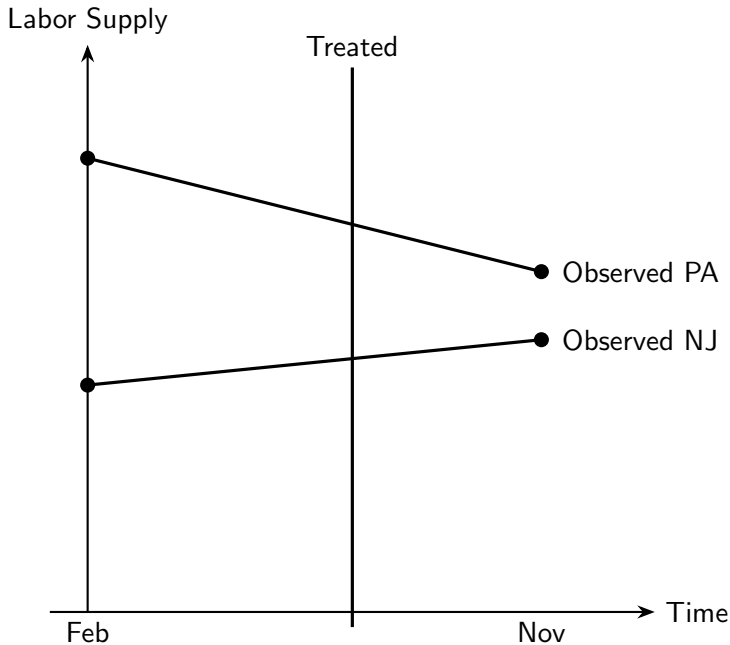
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Formally

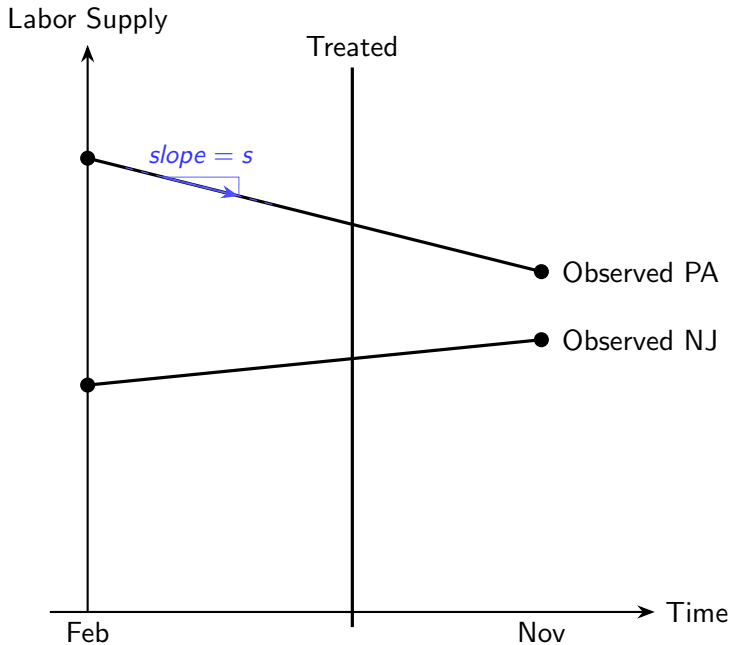
Let $Y_{it}(0)$ = what group g *would have had without treatment* at time t , and $\Delta Y_i(0) = Y_{i,\text{post}}(0) - Y_{i,\text{pre}}(0)$.
Then:

$$\Delta Y_T(0) = \Delta Y_C(0)$$

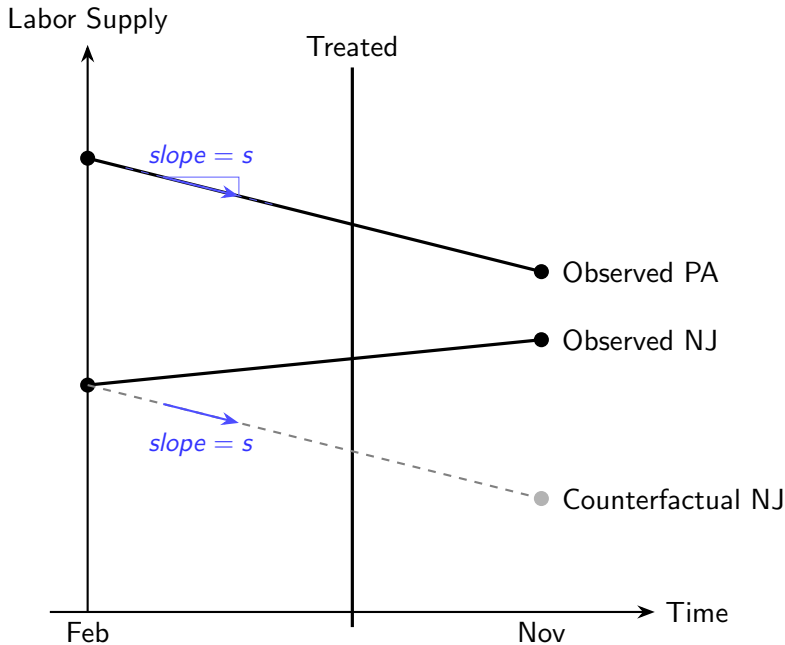
DiD: Parallel Trends Graphically



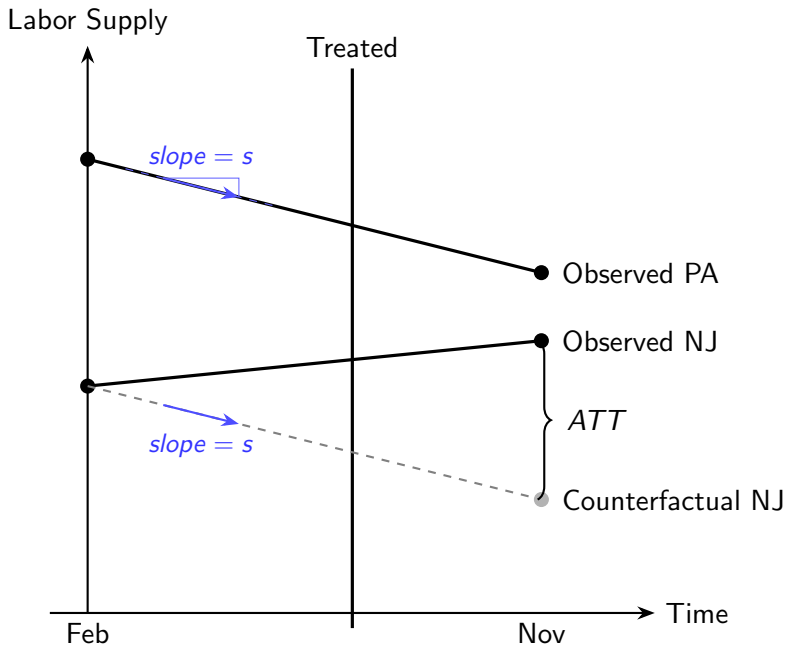
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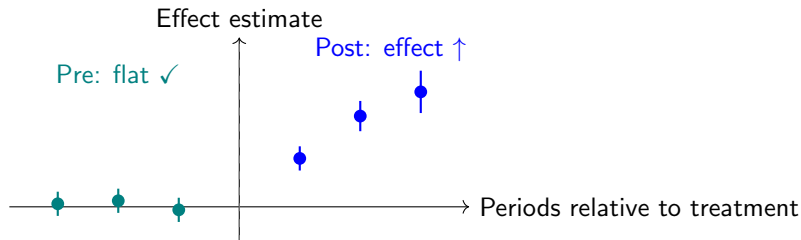
DiD: Checking Parallel Trends

Parallel trends is untestable — we never observe the untreated counterfactual after treatment.

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In practice: Use an **event study plot** to examine pre-treatment trends.

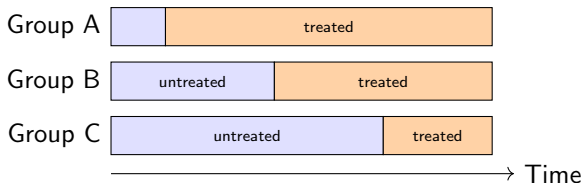


Flat pre-treatment coefficients **support** (but cannot prove) parallel trends.

DiD: Modern Extensions

Real-world data is rarely as simple as two groups over two periods.

- ▶ **Staggered Treatment Adoption** (policies roll out at different times):

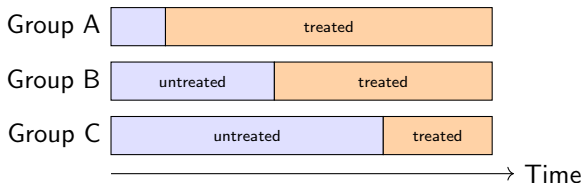


Strategy: Use **Callaway & Sant'Anna (2021)** or **Sun & Abraham (2021)** to estimate clean group-time treatment effects.

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Strategy: Use **Callaway & Sant'Anna (2021)** or **Sun & Abraham (2021)** to estimate clean group-time treatment effects.

- ▶ **No Perfect Control Group:**
 - ▶ **Triple Difference (DDD):** Use a third group to subtract out confounding local trends.
 - ▶ **Synthetic Control:** Blend control units to match the treated unit's pre-trend.
 - ▶ **Sensitivity analysis:** Bound the ATT under varying parallel trends violations.

DiD: Key Takeaways

When to use DiD

- ▶ You have **repeated observations** over time.
- ▶ A credible **control group** exists with similar pre-trends.
- ▶ A policy affects one group but not the other.

Limitations

- ▶ Parallel trends is **untestable** post-treatment.
- ▶ Need to have no **anticipatory behavior**
- ▶ Spillovers between treated and control units can bias the estimate.

DiD gives us the ATT: the average causal effect of a policy on those who received it, provided parallel trends holds.

Exercise

Background

In my Kingdom:

- ▶ The King's cousin loves candies, so his dental health becomes terrible.
- ▶ In 2020, the King proposed **The Great Candy Ban**.
- ▶ But — only half of the provinces implement it.
- ▶ Did the Candy Ban actually improve children's dental health?

Policy Setup

```
▶ # Parameters
n_provinces = 20
years = np.arange(2015, 2024)

data = []

for province in range(n_provinces):

    # First 10 provinces obey the King
    treated = 1 if province < 10 else 0

    for year in years:

        # Post period: after sugar ban in 2020
        post = 1 if year >= 2020 else 0

        # Baseline dental health score
        base = 60

        # General upward trend (dentistry improving over time)
        trend = 0.2 * (year - 2015)

        # Treatment effect: sugar ban improves dental health
        treatment_effect = 3 if (treated == 1 and post == 1) else 0

        # Random noise
        noise = np.random.normal(0, 0.9)

        dental_health = base + trend + treatment_effect + noise
        data.append([province, year, treated, post, dental_health])

df = pd.DataFrame(data, columns=["province", "year", "treated", "post", "dental_health"])
```

Dataset Structure

Panel Dataset (2015–2023)

- ▶ province : Province ID
- ▶ year : Year
- ▶ treated : Candy ban
province (=1)
- ▶ post : After 2020 (=1)
- ▶ dental_health : Outcome
variable

	province	year	treated	post	dental_health
0	0	2015	1	0	60.563101
1	0	2016	1	0	59.428558
2	0	2017	1	0	59.436197
3	0	2018	1	0	61.034225
4	0	2019	1	0	60.598883
...	...	...	...	...	...
175	19	2019	0	0	59.897724
176	19	2020	0	1	60.983338
177	19	2021	0	1	60.940207
178	19	2022	0	1	61.690447
179	19	2023	0	1	60.855492

180 rows × 5 columns

Why Parallel Trends Matter

Before we trust a DiD estimate, we must assume:

Parallel Trends Assumption

In the absence of treatment, the treated group would have followed the same trend as the control group.

What Does the Code Do?

This code computes:

- ▶ Average dental health
- ▶ For each year
- ▶ Separately for treated and untreated provinces

```
# Calculate average dental health by year and treatment group
avg = df.groupby(["year", "treated"])["dental_health"].mean().reset_index()

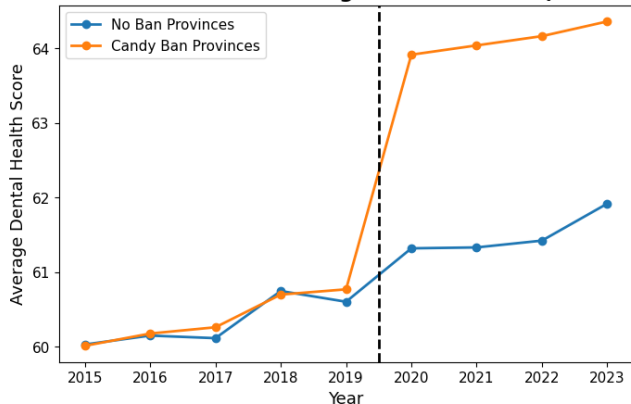
plt.figure(figsize=(8,5))

for t in [0,1]:
    subset = avg[avg["treated"]==t]
    label = "Sugar Ban Provinces" if t==1 else "No Ban Provinces"
    plt.plot(subset["year"], subset["dental_health"],
             marker='o', linewidth=2, label=label)

# Policy line between 2019 and 2020
plt.axvline(x=2019.5, linestyle="--", linewidth=2, color="black")
plt.title("Parallel Trends Check: Average Dental Health (2015-2023)",
         fontsize=16, fontweight="bold")
plt.ylabel("Average Dental Health Score", fontsize=13)
plt.xlabel("Year", fontsize=13)
plt.legend(fontsize=11)
plt.xticks(fontsize=11)
plt.yticks(fontsize=11)
plt.show()
```

Parallel Trends Plot

Parallel Trends Check: Average Dental Health (2015-2023)

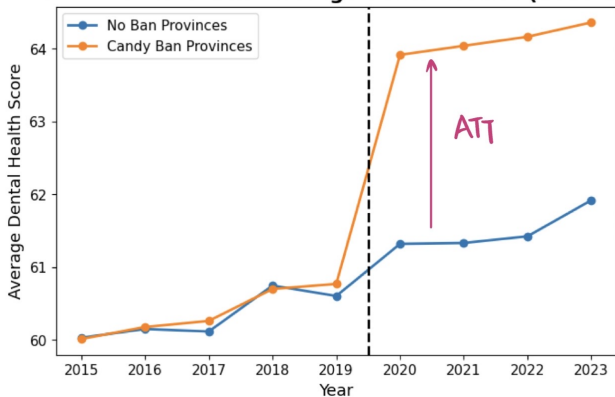


Discussion Questions:

1. What do you see in the plot?

Parallel Trends Plot

Parallel Trends Check: Average Dental Health (2015-2023)



The DiD Regression

We estimate:

$$DentalHealth_{it} = \beta_0 + \beta_1 Treated_i + \beta_2 Post_t + \beta_3 (Treated_i \times Post_t)$$

The key coefficient is β_3 — our DiD estimator.

```
# Create DiD interaction term
df["did"] = df["treated"] * df["post"]

# Run Difference-in-Differences regression
model = smf.ols("dental_health ~ treated + post + did", data=df).fit()

print(model.summary())
```

What Does Each Coefficient Mean?

β_0 : Average dental health of the control group before 2020.

β_1 : Baseline difference between treated and control provinces.

β_2 : Common time trend after 2020 (affects both groups).

β_3 : Additional change in treated provinces after 2020.

How to Interpret β_3

β_3 measures:

(Treated After – Treated Before) – (Control After – Control Before)

$\beta_3 \approx 2.6$:

- ▶ Treated provinces improved 2.6 points more than control provinces.
- ▶ After removing general time trends and baseline differences.

OLS Regression Results						
Dep. Variable:	dental_health	R-squared:	0.746			
Model:	OLS	Adj. R-squared:	0.742			
Method:	Least Squares	F-statistic:	172.4			
Date:	Tue, 03 Mar 2026	Prob (F-statistic):	3.67e-52			
Time:	21:39:36	Log-Likelihood:	-231.26			
No. Observations:	180	AIC:	470.5			
Df Residuals:	176	BIC:	483.3			
Df Model:	3					
Covariance Type:	nonrobust					
	coef	std err	t	P> t	[0.025	0.975]
Intercept	60.3297	0.125	482.405	0.000	60.083	60.577
treated	0.0549	0.177	0.310	0.757	-0.294	0.404
post	1.1686	0.188	6.229	0.000	0.798	1.539
did	2.5660	0.265	9.672	0.000	2.042	3.090
Omnibus:	0.294	Durbin-Watson:	2.079			
Prob(Omnibus):	0.863	Jarque-Bera (JB):	0.287			
Skew:	0.094	Prob(JB):	0.866			
Kurtosis:	2.946	Cond. No.	6.57			

- ▶ Next seminar; will be announced via email
- ▶ In future talks, expect to see:
 - ▶ Latest results on causal inference on the Menstrual Products Project
 - ▶ Our new regressions on Homelessness in Toronto
 - ▶ More mini hands on workshops on ML, synthetic controls, DiD, ..

Code



Scan to open the Google Colab notebook

`https://colab.research.google.com/drive/
1wo5uFCHluVgcFzvFNFHUNSdnUnvhZGKF?usp=sharing`

You Are the King

Case 1 — Make the Policy Useless

Even though provinces passed the policy, a black market keeps candy available.

Case 2 — Make It Very Powerful

Candy and soda are both banned.

```
# Treatment effect: sugar ban improves dental health
treatment_effect = 3 if (treated == 1 and post == 1) else 0

#####IF YOU ARE THE KING#####
# 🐘 treatment_effect = 0 if (treated == 1 and post == 1) else 0
# 🦁 treatment_effect = 8 if (treated == 1 and post == 1) else 0
```

You Are the King

Case 3 — Break Parallel Trends (Critical)

Can we still interpret the post-2020 jump as causal?

Case 4 — Add a Common Shock

Does DiD still work?

```
# Random noise
noise = np.random.normal(0, 0.9)

dental_health = base + trend + treatment_effect + noise

#####IF YOU ARE THE KING#####
# pretrend_violation = 0.9 * (year - 2015) if treated == 1 else 0
# dental_health = base + trend + pretrend_violation + treatment_effect + noise

#####IF YOU ARE THE KING#####
# dentist_reform = 2 if year >= 2020 else 0
# dental_health = base + trend + dentist_reform + treatment_effect + noise
```

Challenge Question

We estimate dynamic treatment effects using event time.

- ▶ Event time measures years relative to the policy.
- ▶ One year before the policy (-1) is the reference group.

Now look at the plot and answer:

1. Are the pre-treatment coefficients close to zero?
2. When does the treatment effect begin?
3. Does the effect persist over time?

